



## SQL Fundamental updated Exercises ②

### SQL Aggregates & operators

Q01. `SELECT DISTINCT department  
FROM Students;`

Out put

| department |
|------------|
| IT         |
| HR         |
| Finance    |

Q02. `SELECT department, AVG(age) AS avg-age  
FROM Students  
GROUP BY department;`

Out put

| department | avg-age |
|------------|---------|
| IT         | 20.5    |
| HR         | 22.0    |
| Finance    | 23.0    |

Q03. `SELECT department, COUNT(*) AS Student-Count  
FROM Students  
GROUP BY department  
HAVING COUNT(*) > 1;`

out put

| Department | Student_count |
|------------|---------------|
| IT         | 2             |
| HR         | 2             |

Q04. SELECT Student\_id, name, age, department  
FROM Students  
WHERE age BETWEEN 21 AND 25;

out put

| Student_id | name    | age | department |
|------------|---------|-----|------------|
| 2          | Bob     | 22  | HR         |
| 3          | Charlie | 21  | IT         |
| 4          | Diana   | 23  | Finance    |
| 5          | Eve     | 22  | HR         |

Q05. SELECT student\_id, name, age, department  
FROM Students  
WHERE department IN ('IT', 'HR')  
AND age > 21;

Out Put

| Student_id | name | age | department |
|------------|------|-----|------------|
| 2          | Bob  | 22  | HR         |
| 5          | Eve  | 22  | HR         |

Q6. SELECT department, sum(credits) AS total\_credits  
 FROM courses  
 GROUP BY departments  
 HAVING sum(credits) > 5;

Out Put

| department | total_credits |
|------------|---------------|
| IT         | 11            |

Q7. SELECT Course\_id, Course\_name, department, credits  
 FROM courses  
 WHERE credits < 4;

Out Put

| Course_id | Course_name | Department | Credits |
|-----------|-------------|------------|---------|
| 101       | SQL Basics  | IT         | 3       |
| 104       | EXCEL       | Finance    | 2       |
| 105       | Statistics  | HR         | 3       |

Q8. SELECT Course\_id, Course\_name, credits  
 FROM Course  
 ORDER BY Credits DESC  
 LIMIT 3;

Out Put

| Course_id | Course_name  | Credits |
|-----------|--------------|---------|
| 102       | Python       | 4       |
| 103       | Data Science | 4       |
| 101       | SQL Basic    | 3       |

Q.9. SELECT MAX (grade) AS max\_grade  
 MIN (grade) AS min\_grade  
 AVG (grade) AS avg\_grade  
 FROM enrollments;

output

| max_grade | min_grade | avg_grade |
|-----------|-----------|-----------|
| 90        | 78        | 84.6      |

Q.10. SELECT Course-id, COUNT (\*) AS enrollment-count  
 FROM enrollments  
 GROUP BY Course-id;

out put

| Course-id | enrollment-count |
|-----------|------------------|
| 101       | 1                |
| 102       | 1                |
| 103       | 1                |
| 104       | 1                |
| 105       | 1                |

Q.11. SELECT department,  
 SUM (salary) AS total\_salary,  
 SUM (bonus) AS total\_bonus  
 FROM salaries  
 GROUP BY department;

Out Put

| Department | total salary | total bonus |
|------------|--------------|-------------|
| IT         | 122000       | 18500       |
| HR         | 109000       | 7500        |
| Finance    | 70000        | 6000        |

Q.12. SELECT Department, AVG(salary) AS Avg-Salary  
FROM Salaries  
GROUP BY department  
HAVING AVG(salary) > 55000;

Out Put

| Department | avg-Salary |
|------------|------------|
| IT         | 61000      |
| Finance    | 70000      |

Q.13. SELECT employee-id, name, salary, bonus,  
salary + bonus AS total\_compensation.  
FROM Salaries  
WHERE salary + bonus > 60000;

Out Put

| employee-id | name  | salary | bonus | total_compensation |
|-------------|-------|--------|-------|--------------------|
| 1           | Tom   | 60000  | 5000  | 65000              |
| 3           | Spike | 70000  | 6000  | 76000              |
| 4           | Tyke  | 62000  | 5500  | 67500              |

Q14. SELECT department  
 SUM(budget) AS total-budget,  
 AVG(budget) AS avg-budget  
 FROM projects  
 GROUP BY department  
 HAVING AVG(budget) > 70000;

Out Put

| department | total-budget | avg-budget |
|------------|--------------|------------|
| IT         | 270000       | 135000     |
| Finance    | 80000        | 40000      |

Q15. SELECT project-id, project-name, department,  
 budget  
 FROM projects  
 WHERE budget BETWEEN 5000 AND 120000  
 AND department <> 'Marketing';

Out Put

| PROJECT-id | Project-name   | department | budget. |
|------------|----------------|------------|---------|
| 1          | AI APP         | IT         | 120000  |
| 2          | Payroll system | Finance    | 80000   |
| 5          | HR portal      | HR         | 50000   |